

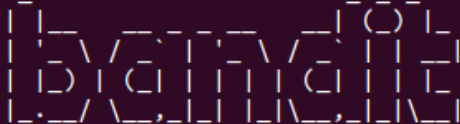
Overthewire [link](#)

You can see the Youtube Playlist to guide you: [Link to Youtube](#)

### Level 0:

```
ssh bandit0@bandit.labs.overthewire.org -p 2220
```

```
sahar@sahar-Vostro-15-3510:~$ ssh bandit0@bandit.labs.overthewire.org -p 2220
```



This is an OverTheWire game server.  
More information on <http://www.overthewire.org/wargames>

backend: gibson-0  
bandit0@bandit.labs.overthewire.org's password:

Password: **bandit0**

### Level 1:

**Is**

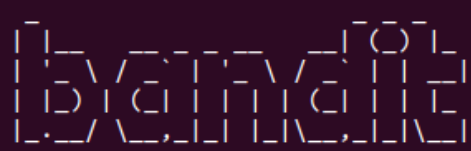
## cat readme

(copy the password)

## exit

```
ssh bandit1@bandit.labs.overthewire.org -p 2220
```

```
bandit0@bandit:~$ ls  
README  
bandit0@bandit:~$ cat README  
Congratulations on your first steps into the bandit game!!  
Please make sure you have read the rules at https://overthewire.org/rules/  
If you are following a course, workshop, walkthrough or other educational activi-  
ty,  
please inform the instructor about the rules as well and encourage them to  
contribute to the OverTheWire community so we can keep these games free!  
  
The password you are looking for is: ZjLjtM6FvvyRnrb2rfNwOZOTa6ip5If  
  
bandit0@bandit:~$ exit  
logout  
Connection to bandit.labs.overthewire.org closed.  
sahar@sahar-Vostro-15-3510:~$ ssh bandit1@bandit.labs.overthewire.org -p 2220
```



```
                This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames
```

backend: gibson-0

bandit1@bandit.labs.overthewire.org's password:

Password: **ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5lf**

Level 2:

ls

cat ./-

exit

ssh bandit2@bandit.labs.overthewire.org -p 2220

```
bandit1@bandit:~$ cat --help
Usage: cat [OPTION]... [FILE]...
Concatenate FILE(s) to standard output.

With no FILE, or when FILE is -, read standard input.

  -A, --show-all           equivalent to -vET
  -b, --number-nonblank     number nonempty output lines, overrides -n
  -e                       equivalent to -vE
  -E, --show-ends          display $ at end of each line
  -n, --number             number all output lines
  -s, --squeeze-blank      suppress repeated empty output lines
  -t                       equivalent to -vT
  -T, --show-tabs          display TAB characters as ^I
  -u                       (ignored)
  -v, --show-nonprinting   use ^ and M- notation, except for LFD and TAB
      --help              display this help and exit
      --version           output version information and exit

Examples:
  cat f - g  Output f's contents, then standard input, then g's contents.
  cat       Copy standard input to standard output.

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Full documentation <https://www.gnu.org/software/coreutils/cat>
or available locally via: info '(coreutils) cat invocation'
```

```
bandit1@bandit:~$ ls
-
bandit1@bandit:~$ cat ./-
263JGJPfgU6LtdEvfgWU1XP5yac29mFx
bandit1@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
sahar@sahar-Vostro-15-3510:~$ ssh bandit2@bandit.labs.overthewire.org -p 2220

      | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ |
      | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
      | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
      | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

      This is an OverTheWire game server.
      More information on http://www.overthewire.org/wargames

      backend: gibson-0
      bandit2@bandit.labs.overthewire.org's password:
```

Password: 263JGJPfgU6LtdEvfgWU1XP5yac29mFx

```
ssh bandit3@bandit.labs.overthewire.org -p 2220
```

[illegible]

Password: **MNk8KNH3Usiio41PRUEoDFPqfxLPISmx**

### Level 4:

## ls -alps

## cd inheres

**ls -al**

## exit

```
ssh bandit4@bandit.labs.overthewire.org -p 2220
```

```
bandit3@bandit:~$ ls -alps
total 24
4 drwxr-xr-x 3 root root 4096 Oct 14 09:26 ./
4 drwxr-xr-x 150 root root 4096 Oct 14 09:29 ../
4 -rw-r--r-- 1 root root 220 Mar 31 2024 .bash_logout
4 -rw-r--r-- 1 root root 3851 Oct 14 09:19 .bashrc
4 drwxr-xr-x 2 root root 4096 Oct 14 09:26 inhere/
4 -rw-r--r-- 1 root root 807 Mar 31 2024 .profile
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -al
total 12
drwxr-xr-x 2 root root 4096 Oct 14 09:26 .
drwxr-xr-x 3 root root 4096 Oct 14 09:26 ../
-rw-r----- 1 bandit4 bandit3 33 Oct 14 09:26 ...Hiding-From-You
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
bandit3@bandit:~/inhere$ exit
logout
Connection to bandit.labs.overthewire.org closed.
sahar@sahar-Vostro-15-3510:~$ ssh bandit4@bandit.labs.overthewire.org -p 2220

  _____
 |  _   _  |  _   _  |  _   _  |  _   _  |  _   _  |  _   _  |
 | |_) | | | |_) | | | |_) | | | |_) | | | |_) | | | |_) | |
 |  _  / |  _  / |  _  / |  _  / |  _  / |  _  / |  _  / |
 | |_) | | | |_) | | | |_) | | | |_) | | | |_) | | | |_) | |
 |  _  / |  _  / |  _  / |  _  / |  _  / |  _  / |  _  / |
 | |_) | | | |_) | | | |_) | | | |_) | | | |_) | | | |_) | |
 |  _  / |  _  / |  _  / |  _  / |  _  / |  _  / |  _  / |

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit4@bandit.labs.overthewire.org's password:
```

Password: **2WmrDFRmJlq3IPxneAaMGhap0pFhF3NJ**

`ls`

- Lists the **names of files and folders** in the current directory.
- Example output:

```
arduino
```

 Copy code

```
file1.txt  folder1  image.png
```

`ls -alps`

Adds several options that make the output much more detailed:

| Option    | Meaning      | Effect                                                            |
|-----------|--------------|-------------------------------------------------------------------|
| <b>-a</b> | all          | shows <b>hidden files</b> (those starting with <code>.</code> )   |
| <b>-l</b> | long listing | shows <b>detailed info</b> (permissions, owner, size, date, etc.) |
| <b>-p</b> | indicator    | adds <code>/</code> after <b>directories</b>                      |
| <b>-s</b> | size         | shows <b>file size in blocks</b> at the beginning                 |

✓ **In short:**

`ls` → simple list

`ls -alps` → **complete detailed view** (hidden files + permissions + sizes + type indicators)



`ls -al`

Adds two useful options to `ls`:

| Option    | Meaning      | Effect                                                                                    |
|-----------|--------------|-------------------------------------------------------------------------------------------|
| <b>-a</b> | all          | shows <b>all files</b> , including hidden ones (those starting with <code>.</code> )      |
| <b>-l</b> | long listing | displays <b>detailed information</b> for each file (permissions, owner, size, date, etc.) |

**Tip: if your terminal is messed up, try the “reset” command.**

Ken tblokelek terminal f level mo3ayan o5roj w awed od5ol just l e5er level solved.

**Tijim ti5dem kil hamasa w ta93ad takiz kima haka:**

```
bandit4@bandit:~$ ls -al
total 24
drwxr-xr-x  3 root root 4096 Oct 14 09:26 .
drwxr-xr-x 150 root root 4096 Oct 14 09:29 ..
-rw-r--r--  1 root root  220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root root 3851 Oct 14 09:19 .bashrc
drwxr-xr-x  2 root root 4096 Oct 14 09:26 inhere
-rw-r--r--  1 root root  807 Mar 31  2024 .profile
bandit4@bandit:~$ cd inhere
```

```
bandit4@bandit:~/inhere$ ls -al
total 48
drwxr-xr-x 2 root root 4096 Oct 14 09:26 .
drwxr-xr-x 3 root root 4096 Oct 14 09:26 ..
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file00
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file01
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file02
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file03
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file04
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file05
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file06
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file07
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file08
-rw-r----- 1 bandit5 bandit4 33 Oct 14 09:26 -file09
bandit4@bandit:~/inhere$ cat -file00
cat: invalid option -- 'f'
Try 'cat --help' for more information.
bandit4@bandit:~/inhere$ cat ./-file00
6);/: d0Jhp000}0k'000vbandit4@bandit:~/inhere$ cat ./-file01
07-(c\0;d0030_~000:#0000b02bandit4@bandit:~/inhere$ cat ./-file02
0Zn0SN:04H000'M3ey0.0-W000 =0bandit4@bandit:~/inhere$ cat ./-file03
0C0000000z0010[0;T
0H000Ts0F000bandit4@bandit:~/inhere$ cat ./-file04
000M0r0203000c00a00S_00000rbandit4@bandit:~/inhere$ cat ./-file05
0h0W0u8000;000l0FP0I0mbandit4@bandit:~/inhere$ cat ./-file06
08000000|30=: B00000S00/0Y["#000bandit4@bandit:~/inhere$ cat ./-file07
40QYVPkxZ00E005pTW81FB8j8lxXGUQw
bandit4@bandit:~/inhere$ exit
logout
Connection to bandit.labs.overthewire.org closed.
sahar@sahar-Vostro-15-3510:~$ ssh bandit5@bandit.labs.overthewire.org
```

[L] [V] [C] [M] [C] [L]  
 [D] [C] [M] [C] [L]  
 [:] [\] [,] [\] [,] [\]

This is an OverTheWire game server.  
More information on <http://www.overthewire.org/wargames>

```
backend: gibson-0
bandit5@bandit.labs.overthewire.org's password:
```

4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw

## Wala te5adem mo5ek w tfiltri

ls -al

cd inhere

ls

find . -type f | xargs file

```
bandit4@bandit:~$ ls -al
total 24
drwxr-xr-x  3 root root 4096 Oct 14 09:26 .
drwxr-xr-x 150 root root 4096 Oct 14 09:29 ..
-rw-r--r--  1 root root  220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root root 3851 Oct 14 09:19 .bashrc
drwxr-xr-x  2 root root 4096 Oct 14 09:26 inhere
-rw-r--r--  1 root root  807 Mar 31  2024 .profile
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls
-file00 -file01 -file02 -file03 -file04 -file05 -file06 -file07 -file08 -file09
bandit4@bandit:~/inhere$ find . -type f | xargs file
./-file02: data
./-file04: data
./-file08: data
./-file00: data
./-file07: ASCII text
./-file06: data
./-file01: data
./-file03: data
./-file05: data
./-file09: data
bandit4@bandit:~/inhere$ cat ./-file07
4oQYVPkxZ00E005pTW81FB8j8lxXGUQw
```

### 1. find . -type f

- `find .` → searches **from the current directory ( . )**
- `-type f` → limits the search to **files only** (not folders)
-  So this part lists all file paths under the current directory and its subdirectories.

### 2. | (pipe)

- Sends ( `pipes` ) the output of the first command ( `find` ) into the next one.

### 3. xargs file

- `xargs` takes the list of file paths from `find` and passes them as arguments to the `file` command.
- `file` identifies the **type of each file** (e.g., "ASCII text", "JPEG image", "ELF executable", etc.)

---

## In short:

This command finds **all files** in the current directory and subdirectories, then displays the **type** of each file.

Level 6:

cd inhere

**find . -type f -size 1033c ! -executable**

cat ./maybehere07/.file2

```
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat ./maybehere07/.file2
HWasnPhtq9AVKe0dmk45nxy20cvUa6EG
bandit5@bandit:~/inhere$
```

HWasnPhtq9AVKe0dmk45nxy20cvUa6EG

1. `find .`  
→ Start searching from the **current directory** ( `.` ) and go through all subdirectories.
2. `-type f`  
→ Look for **files only** (ignore folders, links, etc.).
3. `-size 1033c`  
→ Match files that are **exactly 1033 bytes** in size.
  - `c` = bytes (if you used `k` , it would mean kilobytes; `M` for megabytes, etc.)
4. `! -executable`  
→ The exclamation mark `!` means **NOT**.  
→ So this means **files that are not executable** (you can't run them as programs or scripts).

---

✓ **In short:**

This command finds all **regular files** in the current directory (and below) that:

- are **exactly 1033 bytes** in size, and
- are **not executable**.



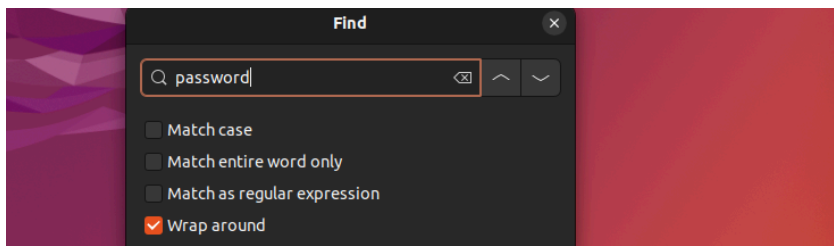
Level 7:

**find / -type f -user bandit7 -group bandit6 -size 33c**

**cat /var/lib/dpkg/info/bandit7.password**

```
bandit6@bandit:~$ find / -type f -user bandit7 -group bandit6 -size 33c
find: '/proc/tty/driver': Permission denied
find: '/proc/1/task/1/fd': Permission denied
find: '/proc/1/task/1/fdinfo': Permission denied
find: '/proc/1/task/1/ns': Permission denied
find: '/proc/1/fd': Permission denied
find: '/proc/1/map_files': Permission denied
find: '/proc/1/fdinfo': Permission denied
find: '/proc/1/ns': Permission denied
find: '/proc/2/task/2/fd': Permission denied
find: '/proc/2/task/2/fdinfo': Permission denied
find: '/proc/2/task/2/ns': Permission denied
find: '/proc/2/fd': Permission denied
find: '/proc/2/map_files': Permission denied
find: '/proc/2/fdinfo': Permission denied
find: '/proc/2/ns': Permission denied
find: '/proc/18/task/18/fdinfo/6': No such file or directory
find: '/proc/18/fdinfo/5': No such file or directory
find: '/lost+found': Permission denied
find: '/var/tmp': Permission denied
find: '/var/spool/bandit24': Permission denied
find: '/var/spool/cron/crontabs': Permission denied
find: '/var/spool/rsyslog': Permission denied
find: '/var/lib/update-notifier/package-data-downloads/partial': Permission denied
find: '/var/lib/apt/lists/partial': Permission denied
find: '/var/lib/polkit-1': Permission denied
find: '/var/lib/chrony': Permission denied
find: '/var/lib/udisks2': Permission denied
find: '/var/lib/snapd/cookie': Permission denied
```

```
find: '/lost+found': Permission denied
find: '/var/tmp': Permission denied
find: '/var/spool/bandit24': Permission denied
find: '/var/spool/cron/crontabs': Permission denied
find: '/var/spool/rsyslog': Permission denied
find: '/var/lib/update-notifier/package-data-downloads/partial': Permission denied
find: '/var/lib/apt/lists/partial': Permission denied
find: '/var/lib/polkit-1': Permission denied
find: '/var/lib/chrony': Permission denied
find: '/var/lib/udisks2': Permission denied
find: '/var/lib/snapd/cookie': Permission denied
find: '/var/lib/snapd/void': Permission denied
find: '/var/lib/private': Permission denied
find: '/var/lib/ubuntu-advantage/apt-esm/var/lib/apt/lists/partial': Permission denied
/var/lib/dpkg/info/bandit7.password
find: '/var/lib/amazon': Permission denied
find: '/var/crash': Permission denied
find: '/var/cache/apt/archives/partial': Permission denied
find: '/var/cache/pollinate': Permission denied
find: '/var/cache/private': Permission denied
find: '/var/cache/apparmor/208b6430.0': Permission denied
find: '/var/cache/apparmor/0fb44ac6.0': Permission denied
```



**morbNTDkSW6jIUc0ymOdMaLnOIFVAaj**



### Level 8:

```
strings data.txt | grep "millionth"
```

```
bandit7@bandit:~$ strings data.txt | grep "millionth"
millionth          dfwvzFQi4mU0wfNbF0e9RoWskMLg7eEc
bandit7@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
sahar@sahar-Vostro-15-3510:~$ ssh bandit8@bandit.labs.overthewire.org -p 2220
```

|   |   |     |   |     |   |     |   |
|---|---|-----|---|-----|---|-----|---|
| 1 | 1 | —   | — | —   | — | (C) | 1 |
| 1 | 1 | \   | / | C   | T | \   | / |
| 1 | 1 | (C) | 1 | (C) | 1 | 1   | 1 |
| 1 | 1 | /   | \ | ,   | 1 | 1   | \ |
| 1 | 1 | \   | / | ,   | 1 | 1   | \ |

This is an OverTheWire game server.  
More information on <http://www.overthewire.org/wargames>

```
backend: gibbon-0
bandit8@bandit.labs.overthewire.org's password:
```

dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc

## Level 9:

```
total 56
4 drwxr-xr-x  2 root    root    4096 Oct 14 09:26 ./
4 drwxr-xr-x 150 root    root    4096 Oct 14 09:29 ../
4 -rw-r--r--  1 root    root     220 Mar 31  2024 .bash_logout
4 -rw-r--r--  1 root    root    3851 Oct 14 09:19 .bashrc
36 -rw-r----- 1 bandit9 bandit8 33033 Oct 14 09:26 data.txt
4 -rw-r--r--  1 root    root     807 Mar 31  2024 .profile
bandit8@bandit:~$ sort data.txt | uniq -c
10 1kopVbuefPoyJ5GDxH0q46SXQVQqZmge
10 1PIVQI2GADz12PtZBMarRD3DpMfDoxbG
10 1S7LmjYT5ALAvRmzK8ksutVRWlMel92r
10 2EFKJco5KZnbkQLj0oen5cVgbwzNioDr
10 2rTFFlUJ4KkcEGhersi6o5QsgofP8yTyr
10 3XiARLa3d6LTJ5PYT5JCFN60VDx6FLVH
1 4CKMh1JI91bUIZZPXDqGanaL4xvAg0JM
10 66TrUQVxbVR6ULKNouQCshyInYXqTYzE
10 6Fv4fN9rP2Jd3789kWRB0aLOLxXTBlo1
10 6GAXZtABu3ubAxFpTGGSNbFm97v20eb1
10 6qkqrxiAscbi9wvRZu0b3ZKuaBKcyBh2
10 6T7s0lWPTNAHZRVJghmVlz9stFDDVvTF
10 6WLU0f8RPUW609z6rdXb2wHufXmbYn8f
10 8qmqiQyJ5rfx3UoDl6ak0FZSmkX78uSy
10 97B26FhKuGsNfM1j6TQGZ3odLa1z7Myb
10 9A0SfySLRMk7os6lcgrhPv0sR3Qfgovn
10 9bFDdDjS8i613oA6Gv5Gv9tyIm6giwie
10 an2aUSb8zZMi4WV9X810UCL54DLCCHqx
10 aYXg6pU5Kn6bzJbU57bB6aLZhGpVs8Nk
10 bTDPuDrLdJm9q3A9yoY81BiLsj3X5kfI
10 C4ciTovIAgxWIumn9oxwq5wqmW0qbBWU
10 CkPCxsHa9CVcFuBPFEDhQviFp2QAj7y6
10 CPL9maTRaiuatGwBVXzJ6MUMZj3unQc
10 d2eVnHqImYkzKbGhUUWU621FPqZVgd2Z
10 dPHudC4CLmhk03aJs7Y7DmGHJra0yW4Q
10 dSQHIMqQ3AZwGwJ7qkgaEXfhjePq0HJg
10 dZ34PbKwxoD1MMFVWMI3Tno4QDgQeRk2
10 EdDaBLSbSMWKSQMGSFvjPXy0jjMReSw3
10 eEAA4ytk957MspwYKLe0gooPzJK0g8jY
10 ExxoxmINoGXLcxaglhG6E8vQvpBnjfFM
10 foTPRYSW9WI0yli5mv03FTRg2C5JB3eU
10 fVMZtRiBkPaERTTtOSZD1gWMXqfS3dp5
10 g5UBLLtZjkLHyPQuCLLDJOa8J7Dvna8M
10 GJLUwQRvWV0TqB9w25ABSq0xzeCSLze3
10 GLEGKhvVtY3fFsYKehdFAEz6SONLtjaQ
10 gmayHpPnu7b0JJtbnZcwVwtjcsLgUSBR
10 gPG2Bto6Kb6tswKkKHCKaLAQ5J1LSWdo
10 GwvgOX51B64aSwgLEyC9BUFEtgEZImA3
10 h0C9mh7wU3MFQv4VUUN0C9MFQv5374h3
```

4CKMh1JI91bUIZZPXDqGanaL4xvAg0JM

# MORE BS



| Command                         | Description                                            | Examples and Options                                   |
|---------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| <i>File Manipulation</i>        |                                                        |                                                        |
| wc                              | Word count.                                            | wc -l mycode.py                                        |
| sort                            | Sort input alphanumerically.                           |                                                        |
|                                 | Pipe/filter output through another program.            | cat foo.txt   less<br>sort myfile.txt   grep mykeyword |
| >                               | Redirect standard output to. Overwrites existing file. | cat foo.txt > foo2.txt                                 |
| >>                              | Redirect standard output and concatenate to.           | cat foo.txt >> foo2.txt                                |
| <                               | Redirect standard input from.                          | less < foo.txt                                         |
| awk                             | Advanced operations by column.                         |                                                        |
| sed                             | Advanced operations by row.                            |                                                        |
| <i>Job/process Manipulation</i> |                                                        |                                                        |
| &                               | Run process in background.                             | ./run_job.sh &                                         |
| jobs                            | Show your current processes                            | jobs                                                   |
| ps                              | Show processes (more flexible)                         | ps -u username                                         |
| top                             | Show top active processes                              | top -u username                                        |
| kill                            | Stop a job                                             | kill processid                                         |
| Ctrl-z                          | Pause current job                                      | ./run_job.sh; Ctrl-z                                   |
| Ctrl-c                          | Cancel current job                                     | ./run_job.sh; Ctrl-c                                   |
| bg                              | Resume process in background                           | ./run_job.sh; Ctrl-z; bg %1                            |
| fg                              | Bring process to foreground                            | fg %2                                                  |

|                           |                                      |                                                                         |
|---------------------------|--------------------------------------|-------------------------------------------------------------------------|
| <i>Remote Connections</i> |                                      |                                                                         |
| ssh                       | Secure shell login.                  | ssh -Y user@host.edu                                                    |
| scp                       | Secure copy.                         | scp user@host.edu:path/to/file.txt .                                    |
| wget                      | Download file from remote URL.       | wget www.google.com -O google.html                                      |
| Ctrl-d                    | Exit login. Sometimes disabled.      | Can also use exit, logout                                               |
| <i>Miscellaneous</i>      |                                      |                                                                         |
| man                       | Manual for other commands.           | man cat                                                                 |
| sleep                     | Do nothing                           | sleep 30                                                                |
| touch                     | Create empty file/update timestamp   | touch newfile.txt                                                       |
| whoami                    | Print current user.                  | whoami                                                                  |
| du                        | Disk usage.                          | du -shc mydirectory/*                                                   |
| df                        | Disk free.                           | df -h                                                                   |
| diff                      | Shows differences between two files. | diff -w file1.txt file2.txt                                             |
| echo                      | Write text to screen.                | echo "Hello? Can you hear me?"                                          |
| which                     | Find path for command.               | which gcc                                                               |
| tar                       | Compress/uncompress files.           | tar -czvf files.tar.gz a.txt b.fits<br>tar -zxvf downloadedstuff.tar.gz |
| history                   | See command history.                 | history   grep myscript.py                                              |
| alias                     | Create command shortcut.             | alias lrt='ls -lhrt'                                                    |
| source                    | Execute file in shell.               | source ~/.bash_profile                                                  |

# SSH (secure shell)



- Used to securely connect to remote systems.
- Encrypts all communication.
- Example: ssh user@server\_ip
- Common use: remote management & file transfer.

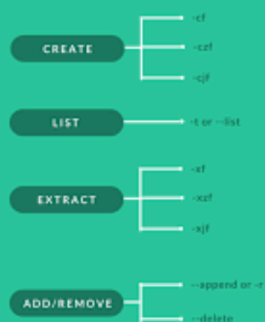


# TAR

- Combines multiple files into one archive.
- Example: tar -cvf archive.tar folder/
- Options:
  - -c create, -v verbose, -f file
  - -x extract files
- Often used for backups and compression.

[Learn More](#) ➔

## Linux tar Command



# SORT

- Sorts lines of text files alphabetically or numerically.
- Example: sort file.txt
- Options:
  - -r → reverse order
  - -n → numeric sort
- Useful for organizing logs or data lists.

[Learn More >](#)



# Git

- Version control system for tracking code changes.
- Commands:
  - git init → start a repo
  - git add . → stage changes
  - git commit -m "message" → save snapshot
  - git push → upload to remote
- Enables teamwork and rollback.

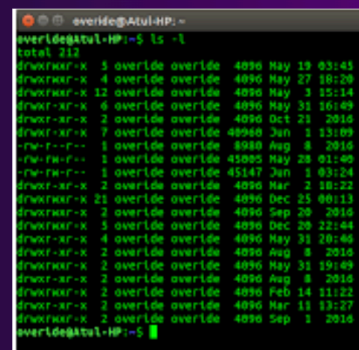
[Learn More >](#)



# SSH SCRIPTS

- A shell script is a text file with a list of Linux commands.
- Used to automate repetitive tasks, setups, and system jobs.
- Interpreted by a shell (like bash or zsh).
- **Make executable with `chmod +x script.sh`.**
- Automates repetitive tasks and setups.

[Learn More >](#)



# INTERPRETERS

- Programs that execute code line by line.
- In Linux, scripts depend on interpreters like bash, python, etc.
- Shebang (`#!/bin/bash`) tells the OS which interpreter to use.
- Ensures consistent execution across systems.

[Learn More >](#)



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